The 1st Forum of the Academy of Medical Sciences of the Serbian Medical Society

The Academy of Medical Sciences of the Serbian Medical Society (Academy) gathers our most famous scientists and experts from various fields of medicine and dentistry. During the 46 years of its activity, the Academy has held over 850 scientific and educational meetings, a large number of lecture cycles for the general population at the Ilija M. Kolarac Endowment, it has published dozens of monographs and thus contributed to the promotion of the scientific and professional work of distinguished members of the Academy, as well as to the education of doctors and the popularization of science.

On March 24, 2023, the Academy will organize the *1st Forum of the Academy of Medical Sciences of the Serbian Medical Society*. It will be a scientific meeting where the research results of the members of the Academy and their coworkers will be presented and discussed with the aim of promoting their scientific work and realizing their better cooperation in subsequent studies. The topic of the *1st Forum of the Academy* is "Prevention, Diagnosis, Treatment and Control of Mass Infectious and Non-Infectious Diseases." It has been our opinion that such a broad topic would attract experts from various fields and thus illustrate the wide-ranging field of scientific interest and work of the Academy members.

Twenty-four papers have been submitted for the *1st Forum of the Academy*. The greatest attention is paid to COVID-19. The results and experiences of the clinical presentation, course, treatment, and outcome of the disease are presented, as well as the results of experimental research and presentations of various post-COVID disorders. These results and experiences are valuable guideposts for planning measures in future epidemics that infectologists and epidemiologists have been warning us about.

In addition to COVID-19, lectures will also be delivered on some important infectious diseases that are not given enough attention, and which can seriously threaten health and life. These are, above all, tuberculosis and fungal diseases, which will be discussed by our well-known experts.

The meeting dedicated to mass diseases could not but include lectures on malignant diseases. Significant results of the study of modern drugs will be presented, as well as research of insufficiently studied tumors that require additional education of doctors, especially concerning the diagnosis of these diseases.

At the Forum, there will be an opportunity to find out the results of clinical and experimental studies on cardiovascular diseases, such as pharmacogenetic studies and analyses of the results of modern methods of cardiovascular diseases treatment.

Studies from the field of dentistry confirm that modern research requires a multidisciplinary and interdisciplinary approach, which has made it possible to achieve exceptional progress in the treatment of oral and dental diseases.

In addition to the presentation from the aforementioned fields, results from ophthalmology, anesthesiology, and otorhinolaryngology will also be presented. The wide range of topics and the quality of submitted abstracts confirm that Academy members contribute significantly to professional and scientific work and the continuous progress of all fields of medicine.

We hope that at the *1st Forum of the Academy*, doctors from different specialties, as well as experts from other related fields, will gather and that a lively discussion will contribute to the quality of the meeting. We wish to all the participants that the Forum fulfills the expectations of us all – the organizers, the authors of the presented studies, and the audience – and that it will be an incentive for the Academy Forum to become a regular annual scientific meeting of the Academy.

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Cardiovascular precision medicine – the role of pharmacogenetic testing

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Introduction/Objective Dual antiplatelet therapy (DAPT) with aspirin plus a $P2Y_{12}R$ inhibitor (clopidogrel or 3rd generation drugs such as ticagrelor or prasugrel) is the standard of care after percutaneous coronary intervention (PCI) to reduce the risk of major adverse cardiovascular events (MACEs). Clopidogrel is a prodrug that requires CYP2C19-catalyzed metabolism to its active form. The gene for the enzyme CYP2C19 is highly polymorphic, so we may distinguish normal metabolizers (NM), intermediate metabolizers (IM) and poor metabolizers (PM). The main objective of our study was to identify IM and PM patients who should be treated with ticagrelor or prasugrel and NM patients, who should receive clopidogrel after PCI for prevention of MACEs.
Methods Using the PCR method in DNA from whole blood of patients after PCI, we analyzed the genotype of 70 patients of average age 66.89 years.
Results The results of our study are as follows: among female patients 71.43% were NM and 28.57% were PM; among male patients 80% were NM, 17.14% were IM and 2.86% (one patient) PM.
Conclusion According to our results that are compatible with genotyping data from other studies on the European population, we may conclude that more than 70% of our patients ≥65 years old are NM and may receive genotype-guided long-term clopidogrel therapy for prevention of MACEs after PCI. This pharmacogenetic testing enables a precision approach in cardiovascular medicine, as for older patients clopidogrel shows a safer profile (lower rate of bleeding events) in comparison with prasugrel and ticagrelor.
Keywords: dual antiplatelet therapy; pharmacogenetic testing; percutaneous coronary intervention; clopidogrel; precision medicine
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